Drug Analysis Project

This project focuses on analyzing drugs, their side effects, and associated medical conditions using exploratory data analysis techniques.

Objective

- To analyze the distribution of drugs across medical conditions
- To identify the most common side effects
- To study prescription patterns (Rx vs OTC)
- To explore drug classes and their clinical significance

```
print("Setup successful")
```

```
Setup successful
```

```
!pip install pandas matplotlib seaborn
```

```
Requirement already satisfied: pandas in c:\users\ramprasad\anaconda3\
lib\site-packages (2.3.3)
Requirement already satisfied: matplotlib in c:\users\ramprasad\
anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: seaborn in c:\users\ramprasad\
anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\ramprasad\
anaconda3\lib\site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
ramprasad\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\ramprasad\
anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\ramprasad\
anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\ramprasad\
anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\ramprasad\
anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\
ramprasad\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\
ramprasad\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\ramprasad\
anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\ramprasad\
anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\ramprasad\
```

```
anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in c:\users\ramprasad\
anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas)
(1.17.0)
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import re

sns.set_style("whitegrid")
```

Dataset Overview

The dataset contains information about drugs, their side effects, medical conditions, drug classes, and prescription types.

```
df = pd.read_csv("drugs_side_effects_drugs_com (2).csv")
df.head()
```

```
        drug_name medical_condition  \
0     doxycycline              Acne
1  spironolactone              Acne
2     minocycline              Acne
3        Accutane              Acne
4     clindamycin              Acne

                                          side_effects
generic_name  \
0  (hives, difficult breathing, swelling in your ...
doxycycline
1  hives ; difficulty breathing; swelling of your...
spironolactone
2  skin rash, fever, swollen glands, flu-like sym...
minocycline
3  problems with your vision or hearing; muscle o...  isotretinoin
(oral)
4  hives ; difficult breathing; swelling of your ...  clindamycin
topical

                                         drug_classes  \
0         Miscellaneous antimalarials, Tetracyclines
1  Aldosterone receptor antagonists, Potassium-sp...
2                                        Tetracyclines
3  Miscellaneous antineoplastics, Miscellaneous u...
4       Topical acne agents, Vaginal anti-infectives

                                          brand_names activity
rx_otc  \
```

```
0  Acticlate, Adoxa CK, Adoxa Pak, Adoxa TT, Alod...      87%     Rx

1                                Aldactone, CaroSpir      82%     Rx

2  Dynacin, Minocin, Minolira, Solodyn, Ximino, V...      48%     Rx

3                                                 NaN      41%     Rx

4  Cleocin T, Clindacin ETZ, Clindacin P, Clindag...      39%     Rx


  pregnancy_category csa alcohol  \
0                  D   N       X
1                  C   N       X
2                  D   N     NaN
3                  X   N       X
4                  B   N     NaN

                                        related_drugs  \
0  amoxicillin: https://www.drugs.com/amoxicillin...
1  amlodipine: https://www.drugs.com/amlodipine.h...
2  amoxicillin: https://www.drugs.com/amoxicillin...
3  doxycycline: https://www.drugs.com/doxycycline...
4  doxycycline: https://www.drugs.com/doxycycline...

                       medical_condition_description  rating
no_of_reviews  \
0  Acne Other names: Acne Vulgaris; Blackheads; B...     6.8
760.0
1  Acne Other names: Acne Vulgaris; Blackheads; B...     7.2
449.0
2  Acne Other names: Acne Vulgaris; Blackheads; B...     5.7
482.0
3  Acne Other names: Acne Vulgaris; Blackheads; B...     7.9
623.0
4  Acne Other names: Acne Vulgaris; Blackheads; B...     7.4
146.0

                                            drug_link  \
0            https://www.drugs.com/doxycycline.html
1          https://www.drugs.com/spironolactone.html
2            https://www.drugs.com/minocycline.html
3               https://www.drugs.com/accutane.html
4  https://www.drugs.com/mtm/clindamycin-topical....

                       medical_condition_url
0  https://www.drugs.com/condition/acne.html
1  https://www.drugs.com/condition/acne.html
2  https://www.drugs.com/condition/acne.html
```

```
3  https://www.drugs.com/condition/acne.html
4  https://www.drugs.com/condition/acne.html
```

```
df.shape
```

```
(2931, 17)
```

```
df.columns
```

```
Index(['drug_name', 'medical_condition', 'side_effects',
'generic_name',
       'drug_classes', 'brand_names', 'activity', 'rx_otc',
       'pregnancy_category', 'csa', 'alcohol', 'related_drugs',
       'medical_condition_description', 'rating', 'no_of_reviews',
'drug_link',
       'medical_condition_url'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2931 entries, 0 to 2930
Data columns (total 17 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   drug_name                       2931 non-null   object
 1   medical_condition               2931 non-null   object
 2   side_effects                    2807 non-null   object
 3   generic_name                    2888 non-null   object
 4   drug_classes                    2849 non-null   object
 5   brand_names                     1718 non-null   object
 6   activity                        2931 non-null   object
 7   rx_otc                          2930 non-null   object
 8   pregnancy_category              2702 non-null   object
 9   csa                             2931 non-null   object
 10  alcohol                         1377 non-null   object
 11  related_drugs                   1462 non-null   object
 12  medical_condition_description  2931 non-null   object
 13  rating                          1586 non-null   float64
 14  no_of_reviews                   1586 non-null   float64
 15  drug_link                       2931 non-null   object
 16  medical_condition_url           2931 non-null   object
dtypes: float64(2), object(15)
memory usage: 389.4+ KB
```

```
df.isnull().sum()
```

```
drug_name                             0
medical_condition                     0
side_effects                        124
generic_name                         43
```

```
drug_classes                        82
brand_names                       1213
activity                             0
rx_otc                               1
pregnancy_category                 229
csa                                  0
alcohol                           1554
related_drugs                     1469
medical_condition_description        0
rating                            1345
no_of_reviews                     1345
drug_link                            0
medical_condition_url                0
dtype: int64
```

Data Cleaning

Missing values were handled by replacing them with appropriate placeholders to ensure consistency in analysis.

```
df['side_effects'] = df['side_effects'].fillna('Unknown')
df['drug_classes'] = df['drug_classes'].fillna('Unknown')
df['rating'] = df['rating'].fillna(0)
df['rx_otc'] = df['rx_otc'].fillna('Unknown')

df.isnull().sum()
```

```
drug_name                           0
medical_condition                   0
side_effects                        0
generic_name                       43
drug_classes                        0
brand_names                      1213
activity                            0
rx_otc                              0
pregnancy_category                229
csa                                 0
alcohol                          1554
related_drugs                    1469
medical_condition_description       0
rating                              0
no_of_reviews                    1345
drug_link                           0
medical_condition_url               0
dtype: int64
```

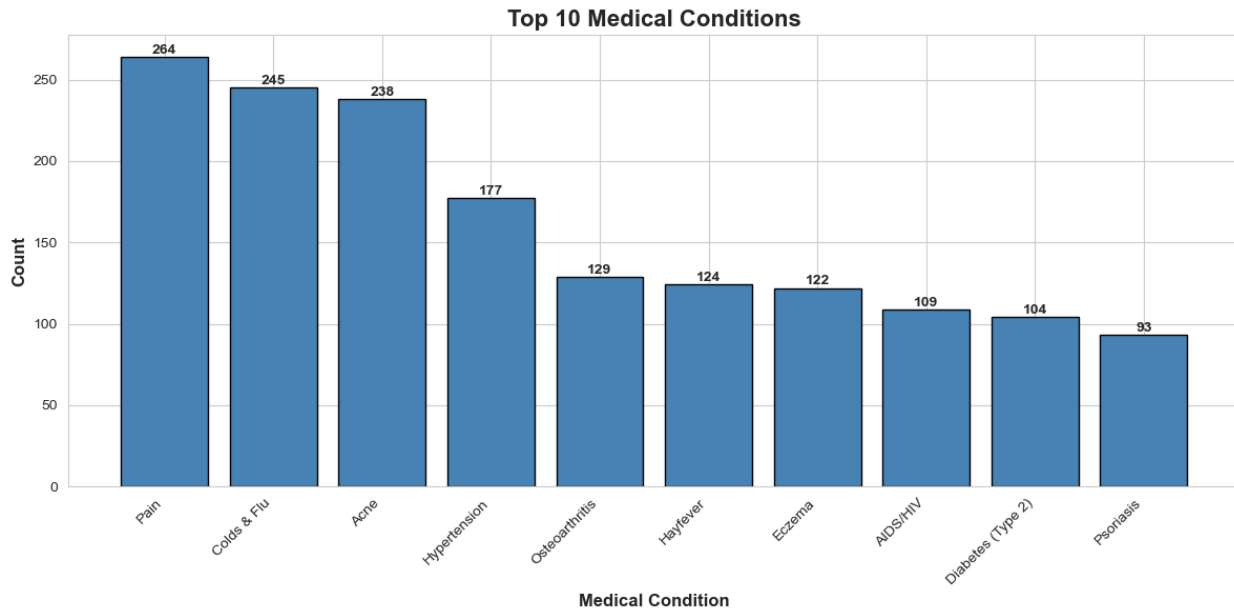
Top Medical Conditions

This section analyzes the most common medical conditions for which drugs are prescribed.

```python
df['medical_condition'].value_counts().head(10)
```

```
medical_condition
Pain                  264
Colds & Flu           245
Acne                  238
Hypertension          177
Osteoarthritis        129
Hayfever              124
Eczema                122
AIDS/HIV              109
Diabetes (Type 2)     104
Psoriasis              93
Name: count, dtype: int64
```

```python
plt.figure(figsize=(12,6))

top_conditions = df['medical_condition'].value_counts().head(10)

bars = plt.bar(top_conditions.index, top_conditions.values,
               color='steelblue', edgecolor='black')

plt.title("Top 10 Medical Conditions", fontsize=16, fontweight='bold')
plt.xlabel("Medical Condition", fontsize=12, fontweight='bold')
plt.ylabel("Count", fontsize=12, fontweight='bold')

plt.xticks(rotation=45, ha='right', fontsize=10)
plt.yticks(fontsize=10)

# Add value labels
for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 2,
             int(bar.get_height()),
             ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



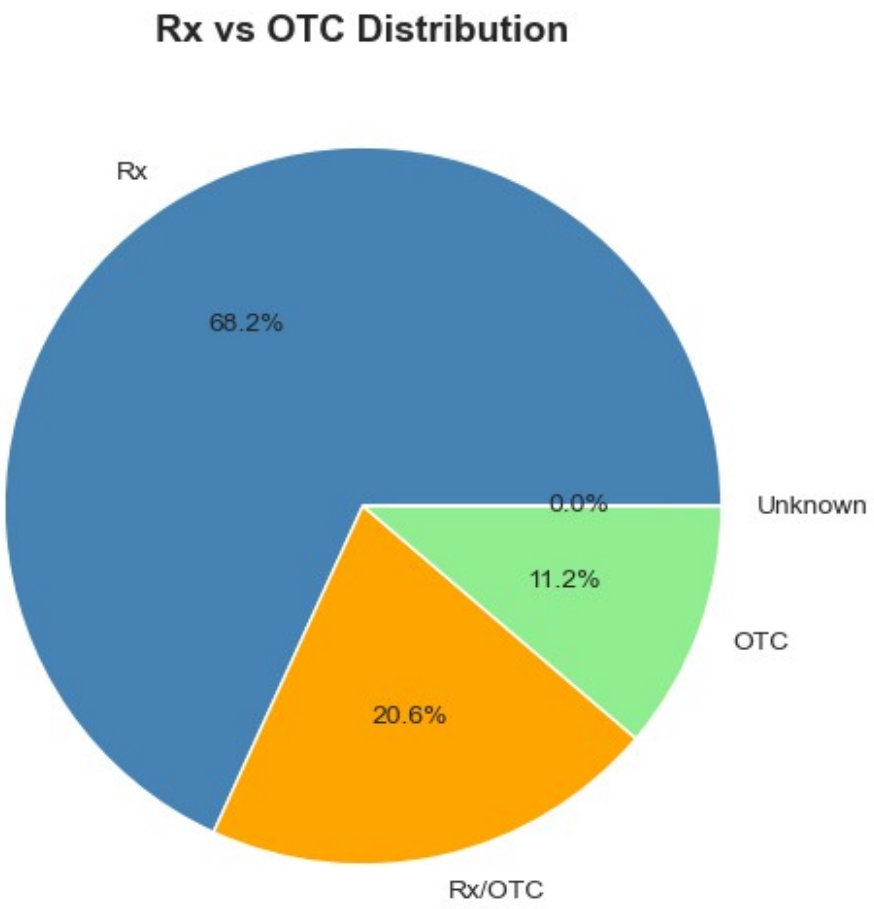
Interpretation

Pain, cold & flu, and acne are among the most commonly treated conditions. This indicates a high prevalence of symptomatic and chronic conditions in clinical practice.

Prescription Pattern Analysis (Rx vs OTC)

This section examines the distribution of prescription and over-the-counter drugs.

```
df['rx_otc'].value_counts()

rx_otc
Rx         1998
Rx/OTC      604
OTC         328
Unknown       1
Name: count, dtype: int64

plt.figure(figsize=(6,6))

df['rx_otc'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    colors=['steelblue','orange','lightgreen']
)

plt.title("Rx vs OTC Distribution", fontsize=14, fontweight='bold')
plt.ylabel("")
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

A majority of drugs fall under prescription category, indicating that many treatments require medical supervision due to potential risks and side effects.

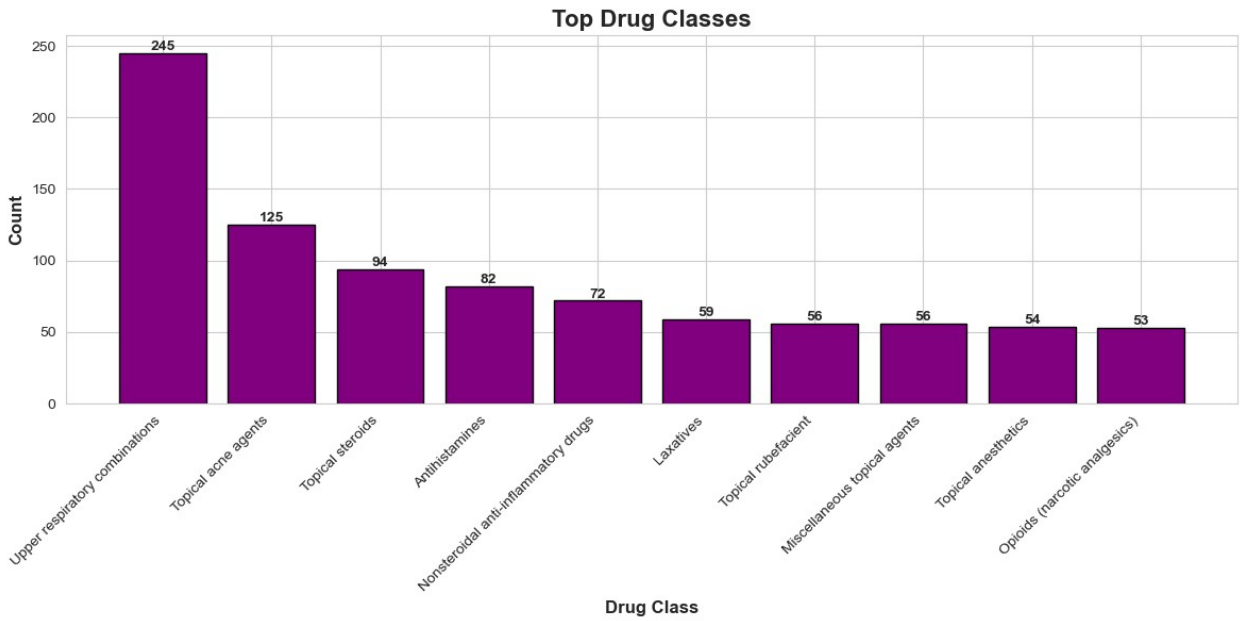
Drug Class Analysis

This section explores the most common drug classes in the dataset.

```
def split_classes(text):
    return re.split(r'[,]', str(text))

classes = df['drug_classes'].apply(split_classes).explode()
classes = classes.str.strip()
classes = classes[classes != "Unknown"]

classes.value_counts().head(10)

drug_classes
Upper respiratory combinations          245
```

```
Topical acne agents                       125
Topical steroids                           94
Antihistamines                             82
Nonsteroidal anti-inflammatory drugs       72
Laxatives                                  59
Topical rubefacient                        56
Miscellaneous topical agents               56
Topical anesthetics                        54
Opioids (narcotic analgesics)              53
Name: count, dtype: int64
```

```
top_classes = classes.value_counts().head(10)

plt.figure(figsize=(12,6))

bars = plt.bar(top_classes.index, top_classes.values,
               color='purple', edgecolor='black')

plt.title("Top Drug Classes", fontsize=16, fontweight='bold')
plt.xlabel("Drug Class", fontsize=12, fontweight='bold')
plt.ylabel("Count", fontsize=12, fontweight='bold')

plt.xticks(rotation=45, ha='right')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 2,
             int(bar.get_height()),
             ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

Drug classes related to respiratory conditions, inflammation, and dermatological treatments are most prevalent, reflecting common therapeutic needs.

Side Effects Analysis

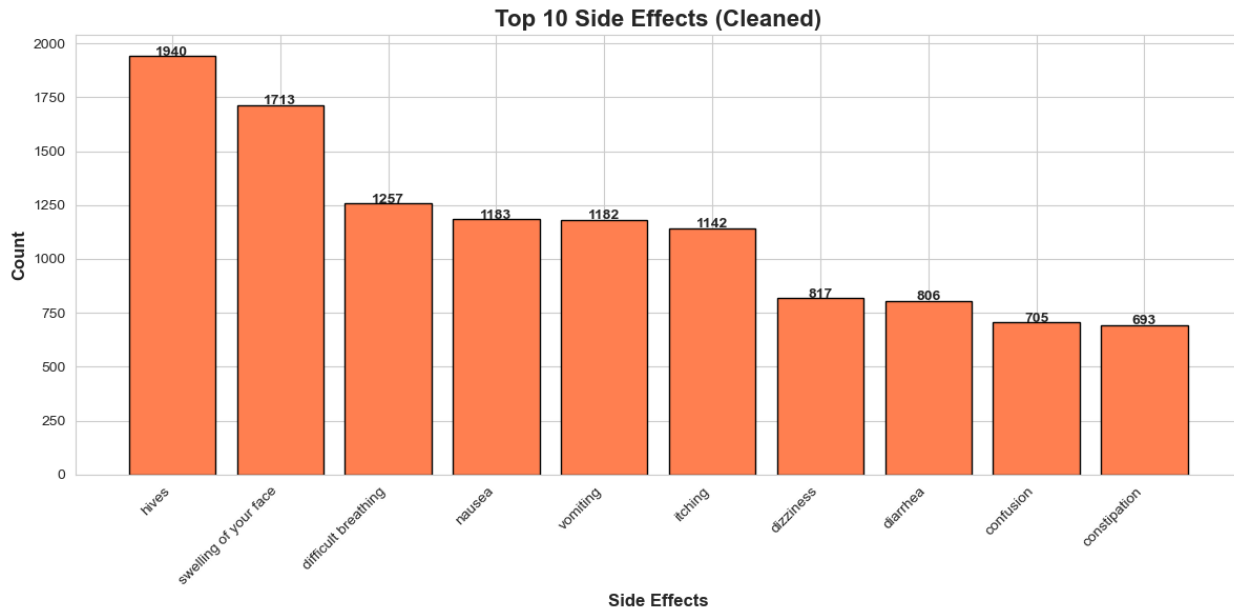
This section identifies the most frequently reported side effects across drugs.

```
# Split side effects
def split_effects(text):
    return re.split(r'[;,]', str(text))

effects = df['side_effects'].apply(split_effects).explode()
effects = effects.str.strip().str.lower()

# Remove empty & unknown
effects = effects[(effects != "") & (effects != "unknown")]

# □ Remove meaningless body parts / noise words
remove_words = [
    'lips', 'tongue', 'face', 'mouth', 'eyes', 'skin'
]

effects = effects[~effects.isin(remove_words)]

# □ Remove very short meaningless words (optional but powerful)
effects = effects[effects.str.len() > 4]

# Get top side effects
top_effects = effects.value_counts().head(10)
```

```python
effects.value_counts().head(10)
```

```
side_effects
lips                      2273
tongue                    2129
hives                     1940
swelling of your face     1713
difficult breathing       1257
nausea                    1183
vomiting                  1182
itching                   1142
dizziness                  817
diarrhea                   806
Name: count, dtype: int64
```

```python
plt.figure(figsize=(12,6))

bars = plt.bar(top_effects.index, top_effects.values,
               color='coral', edgecolor='black')

plt.title("Top 10 Side Effects (Cleaned)", fontsize=16,
fontweight='bold')
plt.xlabel("Side Effects", fontsize=12, fontweight='bold')
plt.ylabel("Count", fontsize=12, fontweight='bold')

plt.xticks(rotation=45, ha='right')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 2,
             int(bar.get_height()),
             ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig("side_effects_cleaned.png", dpi=300)
plt.show()
```



Interpretation

The most common side effects include hives, swelling, and breathing difficulties, indicating a high occurrence of hypersensitivity and allergic reactions associated with drug usage.

Drug Rating Distribution

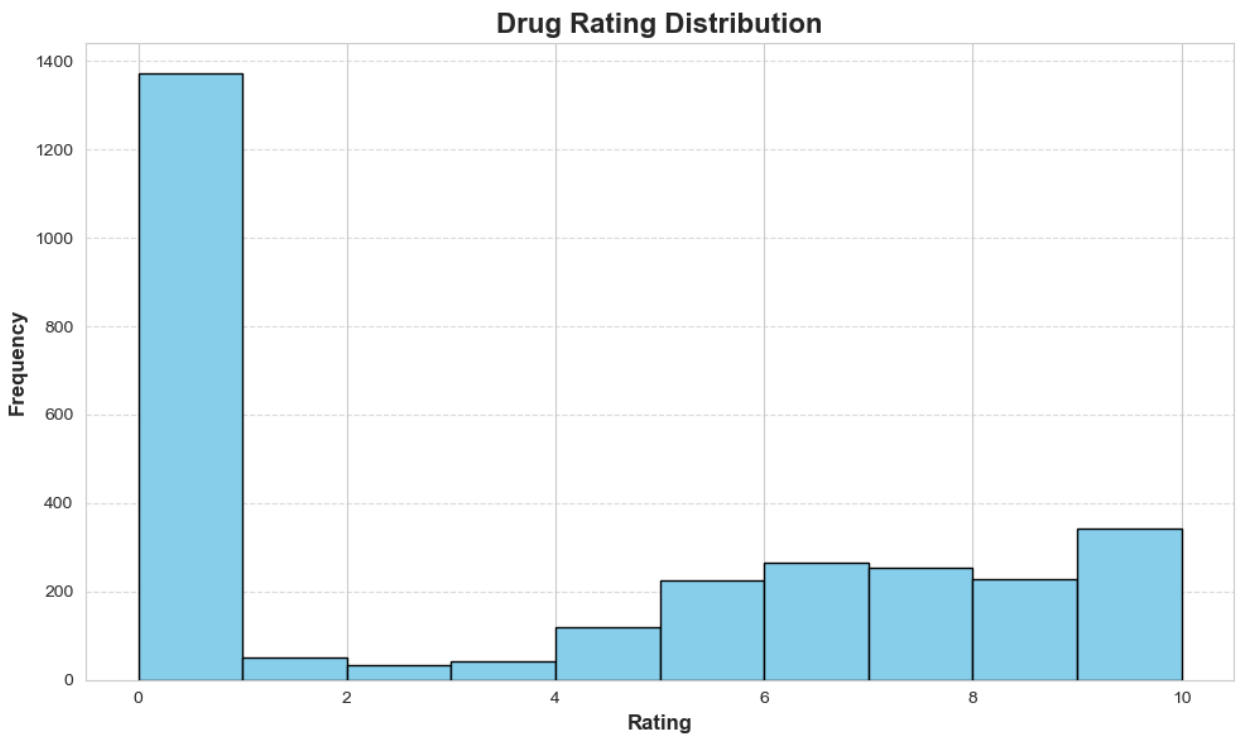
This section analyzes how drug ratings are distributed based on user reviews.

```
df['rating'].describe()
```

```
count    2931.000000
mean        3.686557
std         3.797851
min         0.000000
25%         0.000000
50%         3.300000
75%         7.300000
max        10.000000
Name: rating, dtype: float64
```

```
plt.figure(figsize=(10,6))

plt.hist(df['rating'], bins=10, color='skyblue', edgecolor='black')

plt.title("Drug Rating Distribution", fontsize=16, fontweight='bold')
plt.xlabel("Rating", fontsize=12, fontweight='bold')
plt.ylabel("Frequency", fontsize=12, fontweight='bold')

plt.grid(axis='y', linestyle='--', alpha=0.7)
```

```
plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

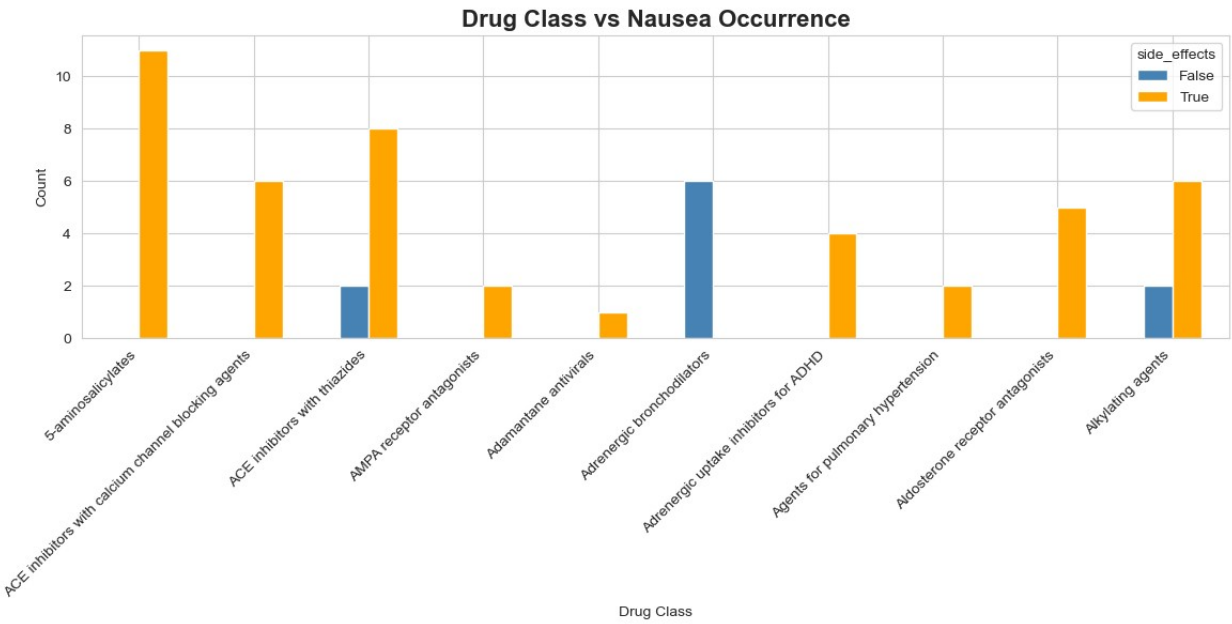
Most drugs have moderate to high ratings, indicating general effectiveness. However, variation in ratings suggests differences in patient response and treatment outcomes.

Side Effects vs Drug Class

This section explores the relationship between drug classes and specific side effects.

```
df['drug_classes_simple'] = df['drug_classes'].str.split(',').str[0]

cross = pd.crosstab(df['drug_classes_simple'],
                    df['side_effects'].str.contains('nausea',
case=False, na=False))

top_cross = cross.head(10)

top_cross.plot(kind='bar', figsize=(12,6),
color=['steelblue','orange'])

plt.title("Drug Class vs Nausea Occurrence", fontsize=16,
fontweight='bold')
```

```
plt.xlabel("Drug Class")
plt.ylabel("Count")

plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

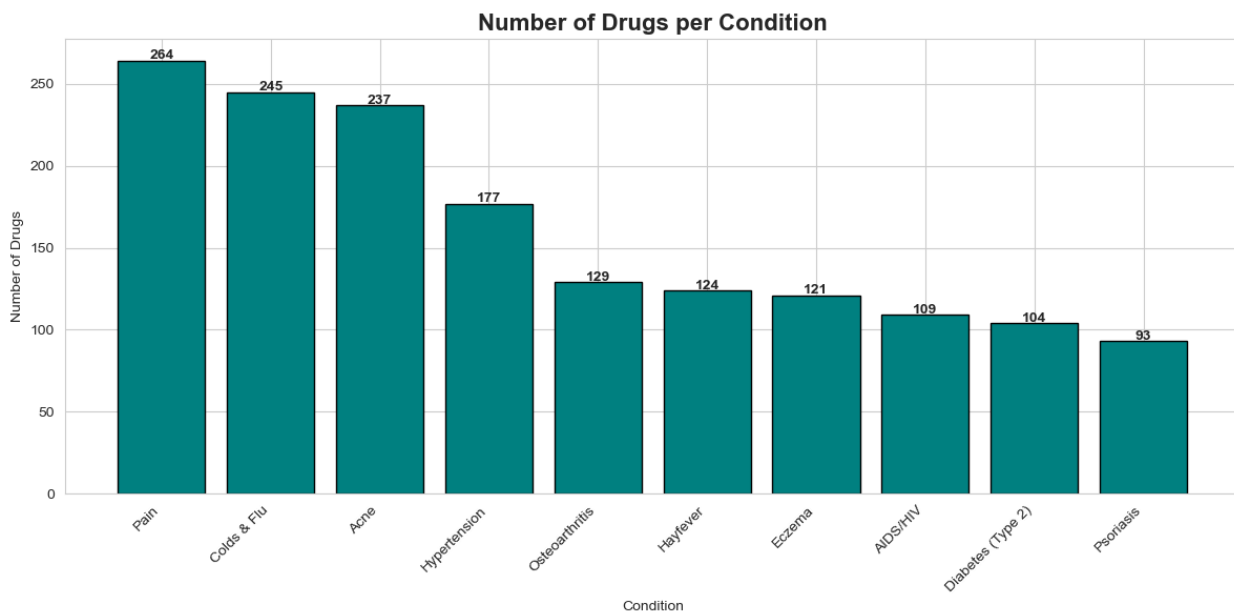
Certain drug classes show higher association with gastrointestinal side effects such as nausea, indicating class-specific adverse effect patterns.

Condition vs Number of Drugs

This section examines how many drugs are available for each medical condition.

```
condition_drugs = df.groupby('medical_condition')
['drug_name'].nunique().sort_values(ascending=False).head(10)

plt.figure(figsize=(12,6))

bars = plt.bar(condition_drugs.index, condition_drugs.values,
               color='teal', edgecolor='black')

plt.title("Number of Drugs per Condition", fontsize=16,
fontweight='bold')
plt.xlabel("Condition")
plt.ylabel("Number of Drugs")
```

```
plt.xticks(rotation=45, ha='right')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height()+1,
             int(bar.get_height()),
             ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

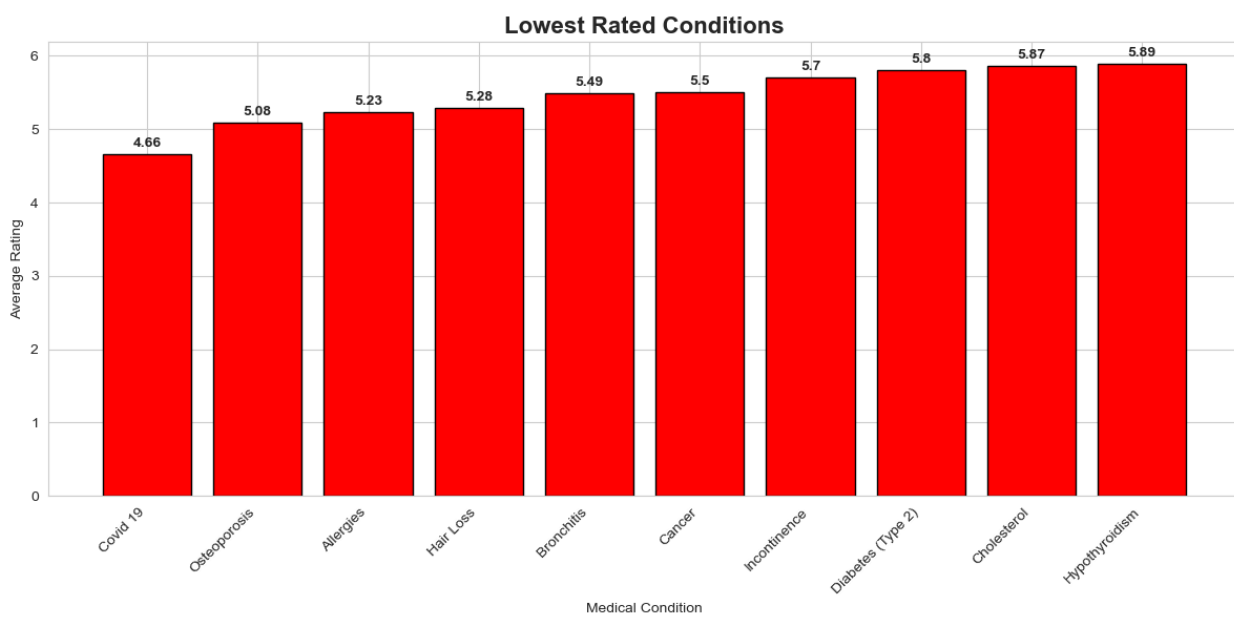
Common conditions such as acne and pain have multiple treatment options, whereas more complex diseases tend to have fewer but specialized drugs.

Rating vs Condition

This section analyzes the average drug rating for each medical condition to evaluate treatment effectiveness across conditions.

```
# Remove invalid ratings (0 = originally missing values)
filtered_df = df[df['rating'] > 0]

# Calculate lowest rated conditions (corrected)
low_rating = filtered_df.groupby('medical_condition')['rating'] \
                        .mean() \
                        .sort_values() \
                        .head(10)
```

```
plt.figure(figsize=(12,6))

bars = plt.bar(low_rating.index, low_rating.values,
               color='red', edgecolor='black')

plt.title("Lowest Rated Conditions", fontsize=16, fontweight='bold')
plt.xlabel("Medical Condition")
plt.ylabel("Average Rating")

plt.xticks(rotation=45, ha='right')

# Add value labels
for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 0.1,
             round(bar.get_height(),2),
             ha='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig("lowest_rated_conditions.png", dpi=300)
plt.show()
```



Interpretation

From a clinical perspective, variation in ratings across conditions suggests differences in drug efficacy, patient tolerance, and disease complexity. Conditions with lower ratings may require improved treatment strategies or personalized therapy approaches.

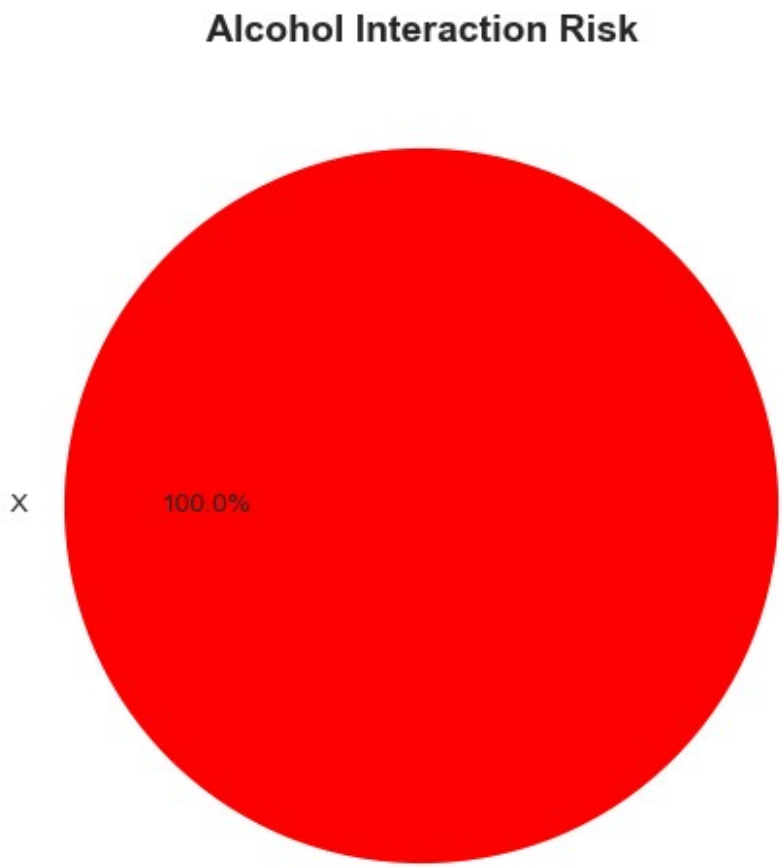
Alcohol Interaction Risk Analysis

This section evaluates how many drugs interact with alcohol.

```
df['alcohol'].value_counts()
```

```
alcohol
X    1377
Name: count, dtype: int64
```

```
plt.figure(figsize=(6,6))

df['alcohol'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    colors=['red','lightgreen']
)

plt.title("Alcohol Interaction Risk", fontsize=14, fontweight='bold')
plt.ylabel("")
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

A considerable number of drugs interact with alcohol,X highlighting the importance of patient counseling to avoid adverse reactions.

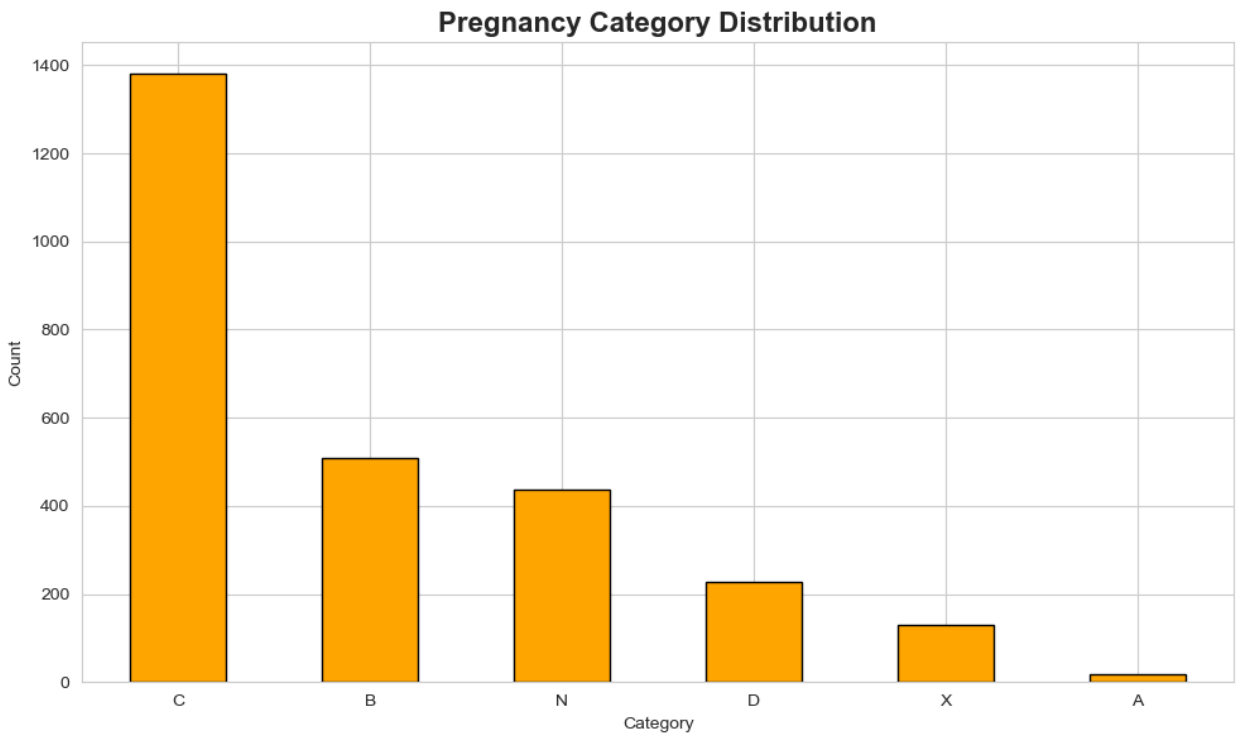
Pregnancy Category Risk Analysis

This section analyzes the safety categories of drugs during pregnancy.

```
df['pregnancy_category'].value_counts()
```

```
pregnancy_category
C    1382
B     509
N     436
D     228
X     129
A      18
Name: count, dtype: int64
```

```
plt.figure(figsize=(10,6))

df['pregnancy_category'].value_counts().plot(
    kind='bar',
    color='orange',
    edgecolor='black'
)

plt.title("Pregnancy Category Distribution", fontsize=16,
fontweight='bold')
plt.xlabel("Category")
plt.ylabel("Count")

plt.xticks(rotation=0)

plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



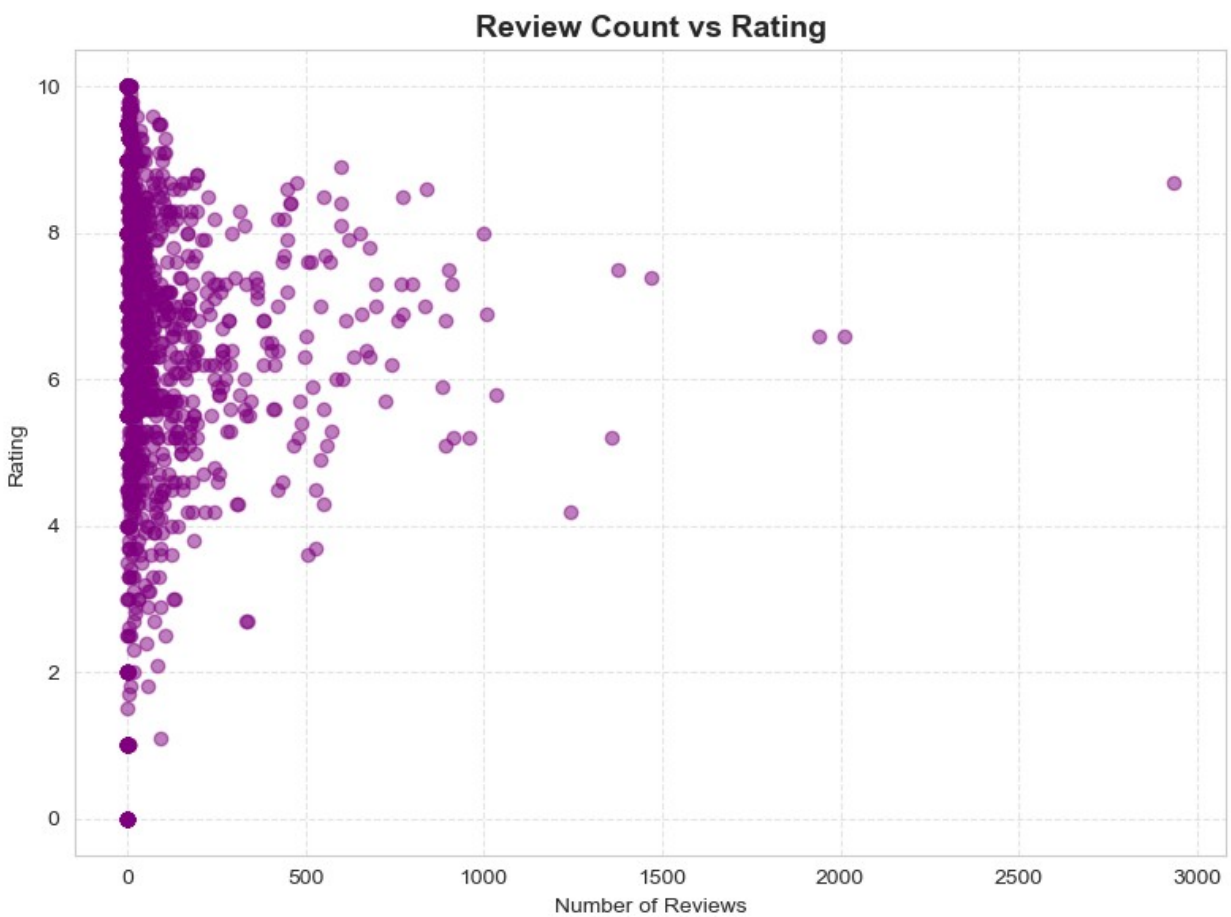
Interpretation

A significant number of drugs fall under category C and D, indicating potential risks and the need for careful risk-benefit evaluation in pregnant patients.

Review Count vs Rating

This section examines the relationship between number of reviews and drug ratings.

```
plt.figure(figsize=(8,6))

plt.scatter(df['no_of_reviews'], df['rating'], alpha=0.5,
color='purple')

plt.title("Review Count vs Rating", fontsize=14, fontweight='bold')
plt.xlabel("Number of Reviews")
plt.ylabel("Rating")

plt.grid(True, linestyle='--', alpha=0.5)

plt.tight_layout()
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



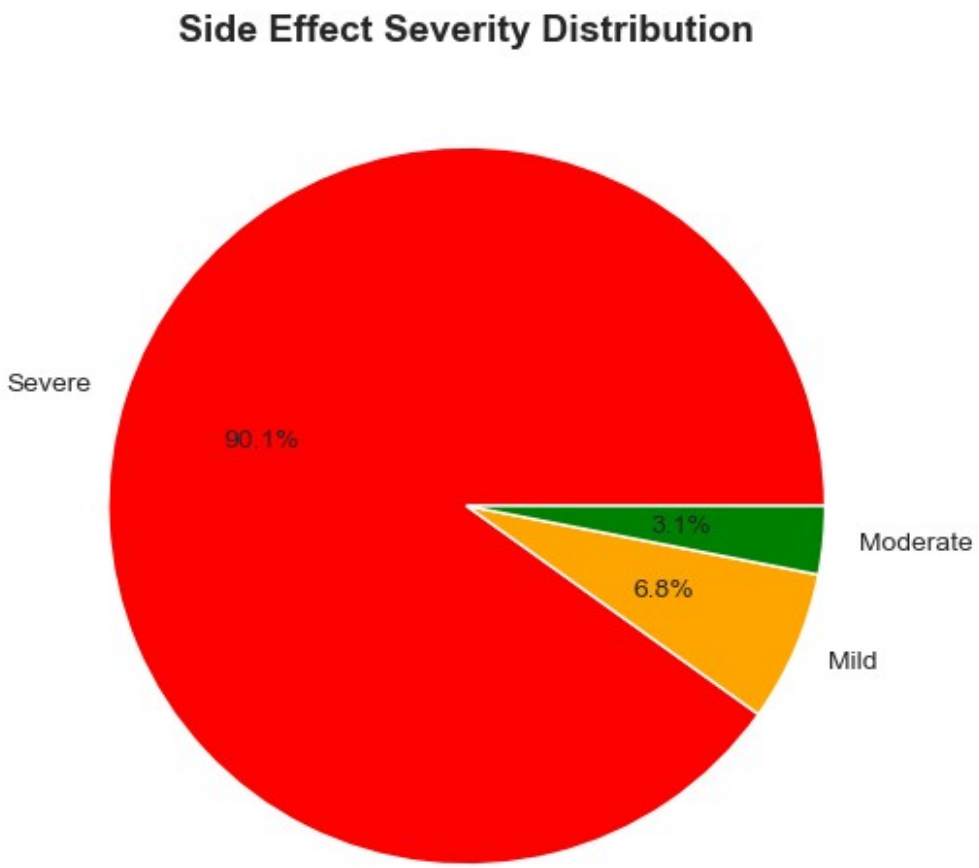
Interpretation

Drugs with a higher number of reviews tend to show more stable ratings, suggesting consistency in effectiveness across larger patient populations.

Side Effect Severity Analysis

This section categorizes side effects based on severity levels.

```
def severity(text):
    text = str(text).lower()
    if 'breathing' in text or 'swelling' in text:
        return 'Severe'
    elif 'dizziness' in text:
        return 'Moderate'
    else:
        return 'Mild'

df['severity'] = df['side_effects'].apply(severity)

severity_counts = df['severity'].value_counts()

plt.figure(figsize=(6,6))

severity_counts.plot(kind='pie', autopct='%1.1f%%',
                     colors=['red','orange','green'])

plt.title("Side Effect Severity Distribution", fontsize=14,
fontweight='bold')
plt.ylabel("")
plt.savefig("top_conditions.png", dpi=300)
plt.show()
```



Interpretation

A notable proportion of drugs are associated with severe side effects such as breathing difficulty and swelling, emphasizing the need for careful monitoring.

Conclusion

This analysis highlights key patterns in drug usage, side effects, and prescription behavior. The findings emphasize the importance of monitoring allergic reactions, understanding prescription trends, and improving patient safety through informed drug use.